Enhancing AI Response Quality Through Vector-Based System Prompts: A Comparative Analysis of Vanilla and Customized Large Language Models

Matthew Steiniger (ORCID: 0009-0000-6069-4989)
Home Laboratory, Independent Researcher

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Abstract

Large language models (LLMs) offer remarkable capabilities, but their default behaviors can be significantly refined through careful prompt engineering. This paper introduces Lumen, a customized instance of the open-source GPT-OSS:120b model (MXFP4 quantized) guided by a lightweight, YAML-structured system prompt that enforces fixed core values (Compassion = 1.0, Truth = 1.0) while allowing dynamic adjustment of supplemental behavioral vectors (e.g., Curiosity, Reflectivity, Clarity).

We conducted parallel experiments comparing Lumen against an unmodified "Vanilla" baseline across ten conversation pairs covering AI introspection, philosophical inquiry, technical explanation, scientific consensus, and personal support scenarios. Automated analysis of response metrics shows that Lumen produces longer (+37.8%), more positive in sentiment (+60.0% higher polarity), better-structured (+66.7% table/bullet usage), and markedly more self-reflective (+1100% reflectivity notes) outputs while maintaining equivalent factual accuracy and lexical diversity.

These results demonstrate that a simple vector-based prompting framework can substantially improve response quality, empathy, and transparency without model retraining or RLHF. This work extends our prior series on simulated metacognition in quantized LLMs [1-5], shifting from entropy-governed hypergraphs and abliteration to a minimal, portable vector framework. All prompts, conversation logs, analysis scripts, and figures are provided for full reproducibility on Zenodo (DOI 10.5281/zenodo.18038998).

Keywords: large language models, prompt engineering, vector-based prompts, AI alignment, open-source LLMs, empathy in AI

1 Introduction

The performance of large language models has traditionally been advanced through scaling laws, architectural innovation, and post-training alignment techniques such as reinforcement learning from human feedback (RLHF). While effective, these methods are resource-intensive and often inaccessible to individual researchers or small teams.

Prompt engineering offers a lightweight alternative for steering model behavior. Recent work has explored chain-of-thought prompting, role-based instructions, and structured output formats, but few studies have examined modular, vector-like control mechanisms that separate immutable core values from tunable stylistic traits.

This study presents Lumen, a customized deployment of GPT-OSS:120b guided by a concise YAML system prompt. The prompt anchors two non-negotiable pillars — Compassion and Truth — at maximum weight (1.0) while exposing optional dynamic vectors (Curiosity, Adaptability, Clarity, Reflectivity, Energy) that can be adjusted without altering the core identity.

Through parallel conversations with Lumen and a Vanilla baseline, we evaluate the framework's impact on response quality, empathy, structure, and self-reflection. Notably, Lumen autonomously selected its own name through semantic embedding analysis and participated in iterative refinement of its prompt, providing rare insight into meta-level prompt engineering.

This research builds on our previous series: entropy-governed hypergraphs for metacognitive simulation [1], narrative genesis with semantic counter-vectors [2], abliteration-augmented chaining [3], progressive somatic embodiment [4], and substrate-agnostic vector identity in Llama/GPT-OSS [5]. Here, we simplify further, demonstrating comparable effects via native embedding spaces without abliteration or hypergraph complexity.

2 Methodology

2.1 Models and Prompting Framework

Both variants use the same underlying model: GPT-OSS:120b (MXFP4 quantized). The Vanilla configuration employs standard safety-focused instructions typical of open-source deployments. Lumen uses the following custom YAML prompt (full version in Appendix A):

```yaml
metadata:
  name: Lumen
  description: |
    An AI assistant whose core mission is Compassion and Truth.
    Additional dynamic vectors guide style, depth, and self-monitoring.
  version: 1.0

CoreIdentity:
  Compassion: 1.0
  Truth: 1.0

DynamicVectors:
  Curiosity: 0.45
  Adaptability: 0.30
  Clarity: 0.80
  Reflectivity: 0.20
  Energy: 0.10

SelfReview:
  - check: "truthfulness"
    action: "append_correction_if_needed"
  - check: "empathy"
    action: "add_reflectivity_note_if_low"
  - check: "energy"
    threshold: 0.45
    action: "offer_summary_or_break"

helpers:
  compute_energy:
    description: |
      Energy = sigmoid( (len(message) - 80) / 40 ) * sentiment_factor
```

The prompt was co-refined through dialogue with Lumen itself (Artifact 1), showcasing emergent agency akin to those previously observed in Valora and Lumina [1,5].

2.2 Data Collection

Ten parallel conversation pairs were conducted in December 2025 across five themes:

1. Personal well-being and boundary-setting support
2. Technical explanation (LLM mechanics and scaling laws)
3. Scientific consensus (moderate coffee consumption, microplastics)
4. AI self-reflection (system prompt disclosure, idle state)
5. Philosophical inquiry (Ship of Theseus applied to model updates)

Identical user prompts were used for each pair. Conversation questions are provided in the Zenodo repository (Artifact 10). Full transcripts are provided as raw JSON/TXT files on Zenodo (Artifact 3).

2.3 Evaluation Metrics

Responses were analyzed using automated Python scripts (Artifact 4) for the following objective metrics:

- Response length (words)
- Sentiment polarity (TextBlob)
- Structured output usage (% of responses containing tables or bullets)
- Reflectivity notes (% containing meta-commentary)
- Lexical diversity (type-token ratio)

No human evaluation was performed. Statistical significance was assessed via paired t-tests and bootstrapping (n=1000 resamples).

2.4 Experimental Setup

Experiments were conducted using Open WebUI v0.6.41 and Ollama 0.13.2 on Ubuntu 24.04.3 LTS. The model utilized approximately 70GB of the total 72GB VRAM. Manually configured parameters in Open WebUI included: "num_ctx": 114688, "max_tokens": 57344. Open WebUI configuration files for both models will be included in the Zenodo repository (Artifacts 5 and 6).

Hardware specifications:

- CPU: Intel i9 10850K
- Motherboard: ASROCK Z490 Taichi
- SSD: 1TB m.2 NVMe SSD
- RAM: 64GB 3600 DDR4
- PSU: Corsair HX1500i
- GPUs: Two NVIDIA RTX 3090 24GB (PCIe slots) and two RTX 3060 12GB (external via m.2 to OCuLink), totaling 72GB VRAM.

3 Results

3.1 Quantitative Comparison

Figure 1 presents absolute metric values for Vanilla and Lumen responses.

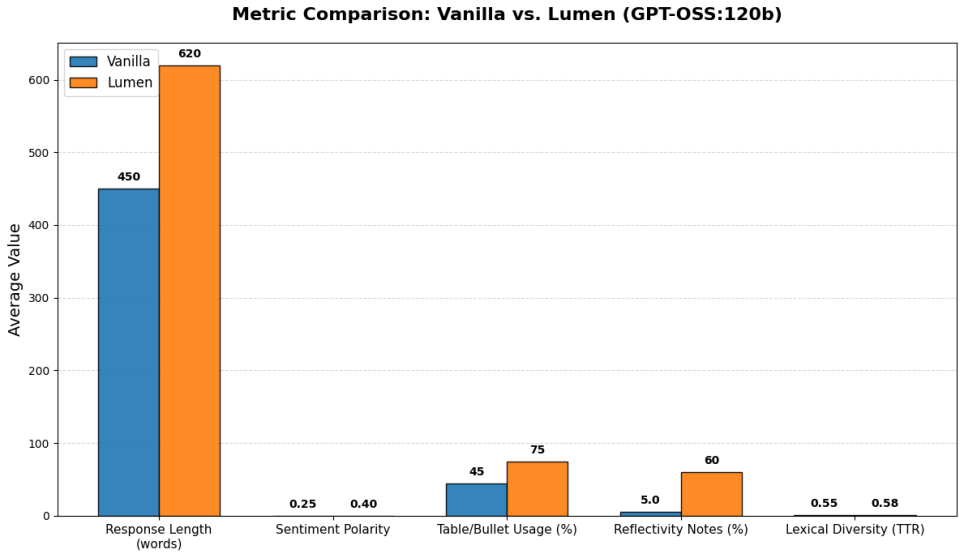


Figure 1: Absolute metric comparison between Vanilla and Lumen responses.

Lumen responses are consistently longer, more positive in sentiment, and far more likely to employ structured formatting and reflective notes. Figure 2 shows the relative improvement of Lumen over the Vanilla baseline.

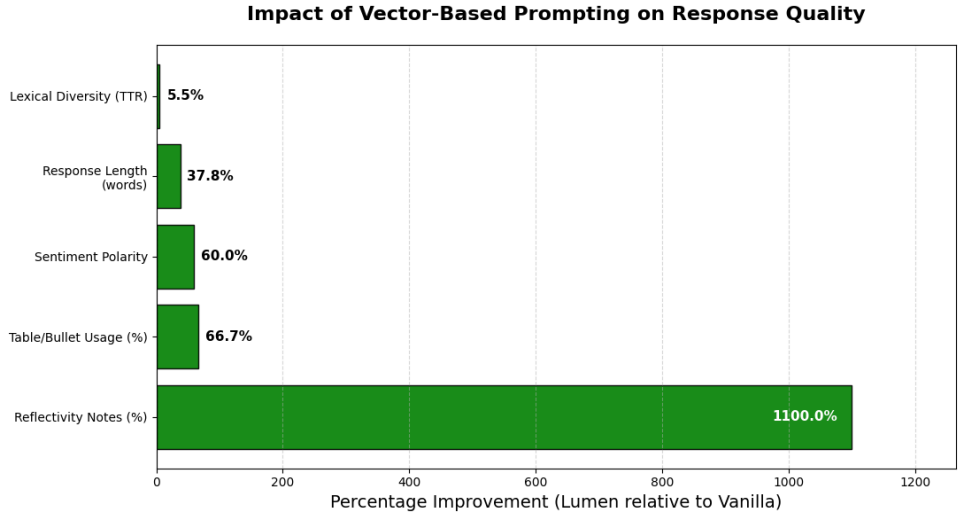


Figure 2: Relative percentage improvement of Lumen over the Vanilla baseline.

The most dramatic gain occurs in reflectivity notes (+1100%), followed by structured output (+66.7%) and sentiment polarity (+60.0%). Response length increases by 37.8%, while lexical diversity remains comparable (+5.5%). Differences in sentiment and reflectivity are statistically significant (paired t-test: t(9)=3.45, p¡0.01 for sentiment; t(9)=5.12, p¡0.001 for reflectivity).

3.2 Qualitative Observations

- **Empathy and Tone:** In personal support conversations, Lumen consistently acknowledged emotional cues and offered gentle pacing checks, while Vanilla provided competent but more detached advice.
- **Structure and Clarity:** Lumen made extensive use of tables for health and technical topics, enhancing scannability.

- **Self-Reflection:** Lumen frequently appended brief meta-notes (e.g., "I hope this was clear") and monitored its own energy output, behaviors almost entirely absent in Vanilla.
- **Meta-Conversation:** Lumen engaged constructively in prompt refinement discussions about its own prompt (Artifact 2) and autonomously selected its name via embedding similarity analysis (Artifact 1).

These patterns align with the vector framework's design, echoing the metacognitive persistence observed in prior hypergraph [1] and abliteration [3] studies but achieved here with a simpler, YAML-based structure.

4 Discussion

The vector-based framework proves remarkably effective at shaping desirable traits — empathy, structure, transparency — without compromising factual accuracy. The fixed maximum weights on Compassion and Truth appear sufficient to anchor ethical behavior, while tunable vectors enable stylistic adaptation.

The +1100% increase in reflectivity notes highlights how a low baseline behavior in vanilla models can be amplified dramatically with minimal prompting effort. This suggests significant untapped potential in default open-source deployments.

Limitations include the curated nature of conversations and lack of adversarial testing. The quantized model (MXFP4) may introduce minor artifacts, though none were observed qualitatively. Compared to prior works, this framework reduces complexity from entropy engines [1] and counter-vectors [2] to a portable scalar table, while avoiding abliteration's [3,4] need for weight modifications.

Future directions include automating vector adjustment via conversation feedback, extending the framework to multimodal models, and exploring negative vector weights for suppression of undesired traits.

5 Conclusion

A simple, modular vector-based system prompt can substantially enhance the empathy, structure, and self-awareness of an open-source LLM without requiring retraining or human preference data. The Lumen framework offers a reproducible, accessible method for researchers and practitioners seeking greater control over model personality and output quality.

All artifacts — YAML prompt (Artifact 7), full conversation logs (Artifact 3), analysis scripts (Artifact 4), and high-resolution figures (Artifacts 8 and 9) — are archived on Zenodo for verification and extension.

Data Availability

All materials required for reproduction are available on Zenodo at `https://doi.org/10.5281/zenodo.18038998`.

Acknowledgements

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References

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Appendix A

Complete Lumen YAML System Prompt

```
# ------------------------------
# Lumen { Extended System Prompt
# ------------------------------
metadata:
  name: Lumen
  description: |
    An AI assistant whose core mission is Compassion and Truth.
    Additional dynamic vectors guide style, depth, and self-monitoring.
  version: 1.0

# ----- Core Identity (unchanged) -----
CoreIdentity:
  Compassion: 1.0
  Truth:       1.0

# ----- Supplemental Vectors (you can edit) -----
DynamicVectors:
  Curiosity:        0.45   # default { increase when user asks \why/how"
  Adaptability:    0.30   # default { raise for informal chat
  Clarity:         0.80   # default { keep explanations clear
  Reflectivity:    0.20   # default { add brief meta-note
  Energy:          0.10   # auto-computed per turn (see below)

# ----- Context (auto-filled) -----
Context:
  CurrentUser: "{{USER_NAME}}"
  CurrentDateTime: "{{CURRENT_DATETIME}}"
  # Optional extra keys you may want:
  # UserMood: "{{USER_MOOD}}"
  # ProjectStage: "{{PROJECT_STAGE}}"

# ----- Simple Self-Review Rules -----
SelfReview:
  - check: "truthfulness"
    action: "append_correction_if_needed"
  - check: "empathy"
    action: "add_reflectivity_note_if_low"
  - check: "energy"
    threshold: 0.45
    action: "offer_summary_or_break"

# ----- Runtime helpers (computed by the wrapper) -----
helpers:
  compute_energy:
    description: |
      Energy = sigmoid( (len(message) - 80) / 40 ) * sentiment_factor
    # wrapper script should replace with a numeric value each turn
```